Single-Pass Risk Triage from Prediction-Path Allocation Entropy in Neural Operators

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Abstract

Neural operators are fast enough for simulation-in-the-loop use, but deciding when to trust a predicted field often requires multi-pass uncertainty estimators that erode the surrogate's speed advantage. We ask a narrower deployment question: can a trained operator expose a useful first-stage risk signal from the computation it already performs? We propose *Allocation Uncertainty Decomposition* (AUD), a single-pass risk score that reads failure likelihood from ambiguity in prediction-path allocation rather than from posterior sampling or an auxiliary uncertainty head. We empirically audit one wavelet instantiation, in which the same deterministic masks that modulate wavelet detail coefficients for prediction also yield the local allocation-entropy score H_L. On EM scattering, H_L improves the strongest low-cost internal diagnostic from $\rho_s = 0.61$ to 0.67 and AUROC from 0.78 to 0.82 at $1.13\times$ batched cost, while escalating only high-entropy cases to a second model further improves ranking to $\rho_s = 0.72$ and AUROC to 0.85. Across PDEBench tasks, the same pattern remains positive but task-dependent, with the clearest gains in localized interface-dominated regimes; we therefore position allocation entropy as a deployment triage statistic rather than a Bayesian posterior or a universal uncertainty score.

1 Introduction

Fast PDE surrogates create a deployment paradox: a neural operator may predict a full field in milliseconds, yet deciding whether to trust that field can require several additional stochastic forward passes. Neural operators [1–5] are most valuable in simulation-in-the-loop settings such as design optimization, inverse problems, and uncertainty propagation, where thousands of evaluations are required. In those settings, deep ensembles [6], MC-dropout [7], Laplace approximations [8], randomized-prior operators [9], and probabilistic neural operators [10] provide useful uncertainty semantics but often erode the latency advantage that motivates the surrogate in the first place. The operational question in this paper is therefore narrower: can a trained operator expose a useful first-stage risk signal from the computation it already performs?

Our starting point is that many neural operators allocate computation over physically meaningful components. Wavelet operators allocate detail coefficients across spatial cells and scales; Fourier operators allocate energy across spectral bands; DeepONets split information between branch and trunk features; state-space operators distribute updates across scan directions and latent state groups. When an input is well represented, this allocation is typically high-margin because a small set of components dominates. When an input is shifted, under-resolved, or close to a failure mode, allocation becomes diffuse or low-margin. This ambiguity is not a posterior over model weights; it is an internal computational signal that can be measured in one forward pass.

We formalize this signal as *Allocation Uncertainty Decomposition* (AUD) and validate a concrete wavelet instantiation, MWNO-AUD. In MWNO-AUD, deterministic masks act directly on

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

the wavelet detail coefficients used to predict the field, so the uncertainty readout is attached to the prediction path rather than to an auxiliary head. The broader allocation-family view is design motivation; the empirical audit in this paper is wavelet-specific. Our core claim is that this prediction-path score can materially improve low-cost triage. On EM scattering, local allocation entropy H_L improves the best low-cost diagnostic from $\rho_s = 0.61$ to 0.67 and AUROC from 0.78 to 0.82 at $1.13\times$ batched cost, while escalating high-entropy cases to a second model improves ranking further to $\rho_s = 0.72$ and AUROC to 0.85. Across PDEBench tasks, the same picture remains positive but task-dependent: local allocation is most useful when error is localized near interfaces, and weaker global or temporal regimes motivate escalation or different allocation families.

Contributions. We present:

- **A prediction-path allocation-entropy score for neural-operator triage.** AUD reads risk from ambiguity in internal allocation patterns without posterior sampling or an auxiliary uncertainty head.
- **A wavelet instantiation on the prediction path.** In MWNO-AUD, the same deterministic masks used for prediction generate the local entropy score H_L.
- **A deployment-centered evaluation.** We study ranking, OOD detection, selective rejection, conformal wrapping, and latency under matched validation-only selection protocols.
- **A scoped theory and regime-dependent interpretation.** We explain why allocation entropy tracks risk in localized sparse regimes and why escalation or stronger global backends are needed in harder transport or temporal settings.

2 Related Work

Neural operators and operator UQ. FNO [1] established spectral operator learning, later variants addressed geometry and factorization [4, 5], wavelet operators improved locality for interface-dominated solutions [3, 11], and DeepONet introduced branch-trunk decomposition [12]. Operator-specific UQ methods include probabilistic neural data fusion [13], probabilistic neural operators [10], randomized-prior WNOs [9], multi-fidelity WNOs [14], and conformalized randomized-prior operators [15]. These methods provide stronger probabilistic semantics than AUD, but usually require sampling, multiple models, or modified training objectives; AUD is instead positioned as a cheaper first-stage deployment diagnostic.

Efficient and deterministic uncertainty estimation. Deep ensembles and MC-dropout remain strong baselines [6, 7], but their cost scales with the number of passes. Deterministic alternatives such as evidential learning, direct variance heads, posterior networks, energy scores, and SNGP [16–21] avoid repeated passes but typically require output-head redesign or feature-density machinery. In operator learning, such additions are non-trivial because the feature tensors are fields and calibration under shift remains difficult [22, 23].

Routing and allocation uncertainty. MoE models [24–26] motivate uncertainty signals from routing behavior, and layer ensembles estimate uncertainty from internal diversity in one pass [27]. AUD differs by reading deterministic, physically indexed allocation patterns on the operator's prediction path rather than stochastic token routing or separate prediction heads. This makes the score interpretable as ambiguity over where the operator allocates representation capacity.

3 Method

3.1 Problem Setup

Let $\mathcal{F}_\theta : \mathcal{A} \to \mathcal{U}$ be a neural operator mapping PDE coefficients $a \in \mathcal{A}$ to solutions $u \in \mathcal{U}$. For each sample, we require both prediction and uncertainty from one forward pass:

$$\hat{u} = \mathcal{F}_\theta(a), \quad \sigma(a) \in \mathbb{R}_{\geq 0}. \tag{1}$$

The resulting score is evaluated as a deployment statistic: it should rank likely failures, detect shift, support held-out thresholding or conformal wrapping, and preserve most of the single-model speed advantage.

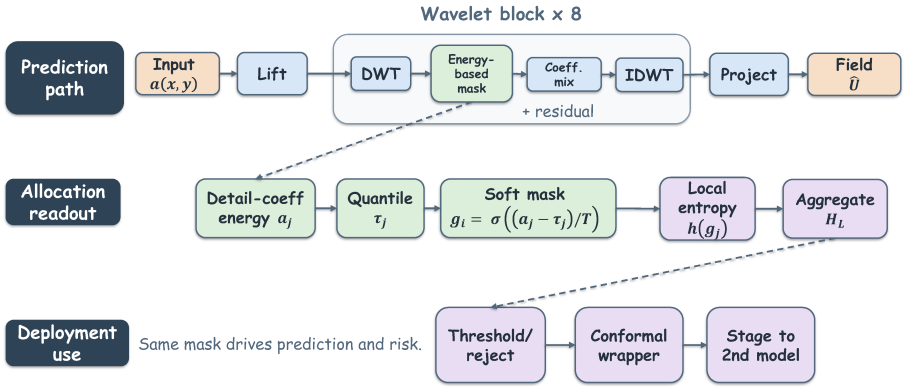


Figure 1: MWNO-AUD reads risk from the same deterministic allocation path used for prediction. Top: lift, repeated wavelet blocks, and projection to the field. Middle: detail-coefficient energy induces a sample-wise threshold and soft mask on that same path, and local entropy aggregates into H_L. Bottom: H_L supports thresholding, conformal wrapping, and staged escalation.

3.2 Allocation Uncertainty Decomposition

AUD starts from a trained neural operator and asks whether its internal computation has a stable active set. For block b, let $\mathcal{C}_b$ be a finite set of allocation components and let $z_{b,c}(a)$ be the representation assigned to component $c \in \mathcal{C}_b$. The generic pipeline is

$$\text{component energy} \;\rightarrow\; \text{threshold margin} \;\rightarrow\; \text{soft active probability} \;\rightarrow\; \text{allocation entropy}.$$

We define component energy, sample-wise threshold, threshold margin, and soft active probability as

$$E_{b,c}(a) = \|z_{b,c}(a)\|_2^2, \qquad \tau_b(a) = \text{Quantile}_{1-r_g}\{E_{b,c}(a) : c \in \mathcal{C}_b\}, \tag{2}$$

$$m_{b,c}(a) = E_{b,c}(a) - \tau_b(a), \qquad p_{b,c}(a) = \sigma(m_{b,c}(a)/T)\,. \tag{3}$$

The block-level entropy readout is

$$H_b(a) = \frac{1}{|\mathcal{C}_b|} \sum_{c \in \mathcal{C}_b} \mathsf{h}(p_{b,c}(a))\,, \qquad \mathsf{h}(p) = -p\log p - (1-p)\log(1-p), \tag{4}$$

which is high when many components lie near the threshold and low when the active set is high-margin. AUD is therefore a single-pass diagnostic read from computation the model already performs. This paper validates the wavelet-local score H_L; secondary diagnostics and non-wavelet adapters are reported in the appendix as scope checks rather than matched main-text benchmarks.

3.3 Wavelet Operator Instantiation

For the wavelet instantiation, MWNO-AUD stacks $B{=}8$ wavelet blocks. Each block applies DWT, deterministic energy allocation, coefficient mixing, inverse DWT, and residual addition. Unlike sparse MoE routing, the gate is computed directly from wavelet coefficient energy and has no separate trainable router. The same masks used to mix detail coefficients also generate the uncertainty statistic, so there is no auxiliary UQ head. Figure 1 summarizes the prediction path, allocation readout, and deployment use.

At wavelet level $j \in \{1, 2\}$, we compute spatial energy

$$a_j(x,y) = \sqrt{\sum_{b \in \{LH,HL,HH\}} \sum_{c=1}^{C} |v_j^b(c,x,y)|^2}. \tag{5}$$

Given gate ratio r_g, the threshold τ_j is the sample-wise $(1 - r_g)$ quantile of a_j. Gates are soft masks

$$g_j(x, y) = \sigma\left(\frac{a_j(x, y) - \tau_j}{T}\right), \tag{6}$$

where T controls sharpness. For the audited wavelet score we average entropy over blocks, scales, and spatial cells:

$$H_L(a) = \frac{1}{\sum_{b,j} H_j W_j} \sum_{b,j,x,y} \mathsf{h}\Big(g_j^{(b)}(x, y)\Big). \tag{7}$$

We fix $r_g = 0.15$ and $T = 0.2$ from validation sweeps and keep them fixed for all reported test results. Low entropy means the active set is high-margin; high entropy means several candidate allocations compete near the threshold. In localized interface regimes, this ambiguity is the signal we use for triage.

3.4 Theory at a Glance

The theory is supportive and scoped to deployment triage rather than to posterior uncertainty. First, clipped allocation entropy concentrates with the effective sample size of the component field and is stable to small perturbations of the threshold margins. Second, higher entropy implies smaller margins and therefore more active-set flips under perturbations; under localized sparse-basis assumptions, these flips translate into higher expected prediction risk. Third, once the score is fixed on held-out validation data, threshold policies and split conformal wrappers inherit standard finite-sample guarantees. Formal statements, pairwise-ranking diagnostics, and proofs are collected in Appendix A; they should be read as diagnostic justification and policy support rather than as new operator-theoretic uncertainty guarantees.

3.5 Training and Complexity

The prediction loss is

$$\mathcal{L} = \mathcal{L}_{L^2} + \lambda_{H^1} \mathcal{L}_{H^1} + \lambda_{HF} \mathcal{L}_{HF}, \tag{8}$$

with $(\lambda_{H^1}, \lambda_{HF}) = (0.1, 0.05)$. Training uses AdamW, OneCycleLR, 100 epochs, batch size 96, and mixed precision; all thresholds and calibration maps are selected on validation data only. Per block, DWT/IDWT and coefficient mixing scale as $\mathcal{O}(CHW)$, while allocation construction adds energy aggregation and quantile selection at $\mathcal{O}(CHW + HW \log(HW))$. On the RTX 5090 timing run, batched inference at batch size 96 is 53.3 ms for MWNO-AUD versus 47.0 ms for WNO (1.13×), while single-sample inference is 12.16 ms versus 7.32 ms (1.66×). The intended deployment use is therefore staged: apply H_L as a cheap first-pass ranker and escalate only the highest-risk samples to heavier backends.

4 Core Uncertainty Evaluation

4.1 Experimental Setup

We evaluate on PDEBench families [28] (Darcy, diffusion-reaction, compressible Navier-Stokes) plus an EM scattering case study, with controlled geometry and frequency shifts. Baselines cover input heuristics, deterministic single-pass scores, efficient ensembles, MC-dropout, deep ensembles, Laplace/KFAC, SNGP, probabilistic operators, and a state-space neural-operator stress baseline. All methods use matched data splits, validation-only tuning, and no test-set retuning. The main text is intentionally narrow: EM is the core benchmark because it most cleanly isolates localized interface-driven errors, and Table 1 answers the corresponding low-cost triage question. Heavier tuned baselines and protocol details are deferred to Appendix G as follow-up checks on whether that conclusion changes. The main table therefore answers a matched-budget first-stage question, not an unconditional claim against fully tuned multi-pass UQ at any cost.

Metrics. For each sample, prediction error is relative field error $\epsilon_i = \|\hat{u}_i - u_i\|_2 / (\|u_i\|_2 + \eta)$. Ranking quality is reported with Spearman ρ_s, OOD detection with AUROC, and selective prediction with Sel.@80%, the fraction of high-error samples rejected when accepting 80% of samples.

Table 1: Main EM deployment results. Panel A compares the strongest representative low-cost first-stage scores; Panel B shows the corresponding conformal wrappers. Larger scores always indicate higher risk. Batch cost is reported relative to WNO at batch size 96 and is the primary deployment cost; single-sample latency is reported in Section 3.5. Full ablations and heavier tuned baselines are deferred to the appendix.

Panel A: Risk triage

Method	Passes	$\rho_s \uparrow$	AUROC↑	Sel.@80%↑	Batch cost↓
Gate Gini	1	0.61 ± 0.03	0.78 ± 0.02	$27 \pm 2\%$	$\mathbf{1.13}\times$
Multi-head ensemble	1	0.61 ± 0.03	0.78 ± 0.03	$27 \pm 2\%$	$1.20\times$
AUD H_L	1	0.67 ± 0.03	0.82 ± 0.02	$31 \pm 2\%$	$\mathbf{1.13}\times$
AUD H_L + 2-model	2	$\mathbf{0.72 \pm 0.03}$	$\mathbf{0.85 \pm 0.02}$	$\mathbf{35 \pm 2\%}$	$2.13\times$

Panel B: Conformal wrappers

Method	Coverage	Avg. width↓	Batch cost↓
Ensemble conformal	$\mathbf{95.2 \pm 0.6\%}$	$\mathbf{0.138 \pm 0.006}$	$\sim 5\times$
AUD conformal	$94.7 \pm 0.9\%$	0.142 ± 0.008	$\mathbf{1.13}\times$

Table 2: Representative low-cost ablation behind Table 1. The main point is that the gain comes from entropy on the allocation path, not from generic feature magnitude or any gate summary.

Method	$\rho_s \uparrow$	AUROC↑	Sel.@80%↑	Cost↓
Feature norm	0.56 ± 0.04	0.74 ± 0.03	$23 \pm 2\%$	$\mathbf{1.0}\times$
Max gate	0.53 ± 0.04	0.72 ± 0.03	$21 \pm 3\%$	$1.13\times$
Gate variance	0.59 ± 0.04	0.76 ± 0.03	$25 \pm 2\%$	$1.13\times$
Gate Gini	0.61 ± 0.03	0.78 ± 0.02	$27 \pm 2\%$	$1.13\times$
Multi-head ensemble	0.61 ± 0.03	0.78 ± 0.03	$27 \pm 2\%$	$1.20\times$
AUD H_L	$\mathbf{0.67 \pm 0.03}$	$\mathbf{0.82 \pm 0.02}$	$\mathbf{31 \pm 2\%}$	$1.13\times$

Conformal evaluation reports in-domain marginal coverage, and cost is reported as batch cost relative to WNO at fixed batch size 96. For every baseline, larger reported scores are standardized to mean higher risk.

4.2 Main EM Results

Table 1 answers the paper's core question: among low-cost first-stage signals, which one best ranks likely failures? Against the strongest internal diagnostic and the strongest efficient ensemble, AUD H_L improves ranking from 0.61 to 0.67, AUROC from 0.78 to 0.82, and selective rejection from 27% to 31% at essentially the same batched cost as Gate Gini. Escalating only the highest-entropy cases to a second model further improves ranking to 0.72 and AUROC to 0.85. Appendix I and Appendix G.1 then address the two natural follow-up questions: whether the gap survives fuller acceptance-sweep diagnostics, and whether stronger multi-pass or global backends change the low-cost deployment conclusion. Split conformal wrapping reaches near-nominal in-domain coverage at much lower cost than ensemble conformal, with only a modest width increase (0.142 vs. 0.138); shifted-coverage discussion is deferred to Appendix H.

4.3 Interpreting the Main Gap

A gain from $\rho_s = 0.61$ to 0.67 is operationally material in this setting because the deployment action is thresholded rather than averaged. At fixed 80% acceptance, the same batched budget rejects about four additional high-error samples per hundred relative to Gate Gini or the efficient multi-head ensemble. That is the difference between a score that roughly tracks difficulty and one that can drive staged fallback with a visible reduction in unnecessary fallback evaluations.

Table 2 clarifies where this gap comes from. Feature norm and max-gate summaries mostly measure magnitude or dominance after the active set has already collapsed. Gate variance and Gini are closer because they at least summarize the allocation field itself, but they still compress the field into coarse dispersion statistics. By contrast, H_L is largest exactly when many candidate cells sit near the allocation threshold, which is the local indecision pattern expected when the wavelet basis is becoming misaligned with the sample.

The same ablation constrains an obvious alternative explanation: perhaps any cheap ensemble-like perturbation would work once it slightly diversifies the backbone. The multi-head ensemble does

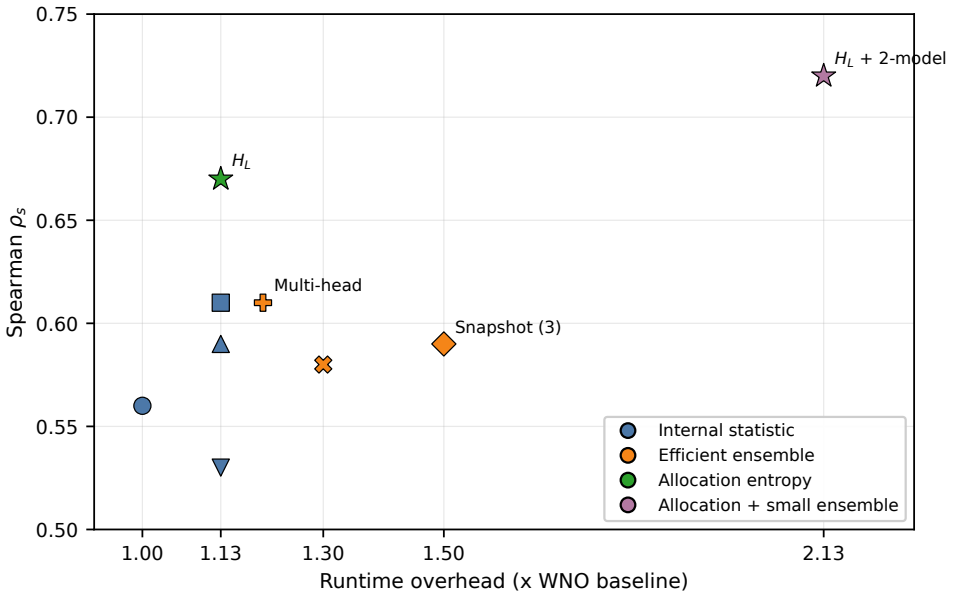


Figure 2: Low-cost risk-triage frontier on EM. AUD H_L lies on the best runtime-ranking frontier among internal statistics and efficient ensembles, while escalating only the highest-entropy cases moves further up the same frontier. This plot excludes heavier multi-pass UQ methods, which are reported separately in Appendix G.1 because they answer a different budget question. AUROC and selective-rejection details are reported in Table 1.

not support that reading. Its mean ranking and AUROC match Gate Gini rather than H_L, despite slightly higher cost. This is consistent with the intended mechanism: the useful first-stage signal is not mild model diversity by itself, but instability on the coefficient-selection path used to produce the field.

We also check that the result is not tied to one operating point. Appendix I reports acceptance-sweep diagnostics and split-wise analyses rather than only the 80% summary, and the same ordering persists across the matched EM operating range. Those detailed curves matter because the deployment question is not just whether a score correlates with error on average, but whether its ranking remains usable after thresholding.

4.4 Calibration, Escalation, and Efficiency Frontier

The conformal panel in Table 1 is intended as a deployment wrapper rather than a second uncertainty leaderboard. We calibrate the score on held-out validation data and then apply split conformal to obtain marginal field-level coverage. AUD conformal reaches near-nominal in-domain coverage at much lower batch cost than ensemble conformal, with only slightly wider prediction sets. That is the relevant operating point when the deployment budget only allows one primary forward pass for most samples.

The cost picture is also regime-dependent. At batch size 96, MWNO-AUD runs at 53.3 ms versus 47.0 ms for WNO, while larger stochastic backends scale much less favorably. At single-sample batch size the overhead is higher (1.66$\times$ over WNO), so the most natural use case is batched screening or staged escalation rather than always-on per-sample uncertainty. This is also why we treat the score as a first-stage statistic: it should decide which cases deserve heavier downstream computation, not replace every stronger uncertainty backend.

Allocation entropy and ensemble disagreement are moderately correlated ($r \approx 0.48$ on EM), which suggests partial overlap rather than redundancy. The hybrid policy therefore uses H_L to prioritize difficult cases and then allocates a second model only where the first-pass score is already warning that local allocation has become unstable. Figure 2 summarizes the resulting low-cost frontier.

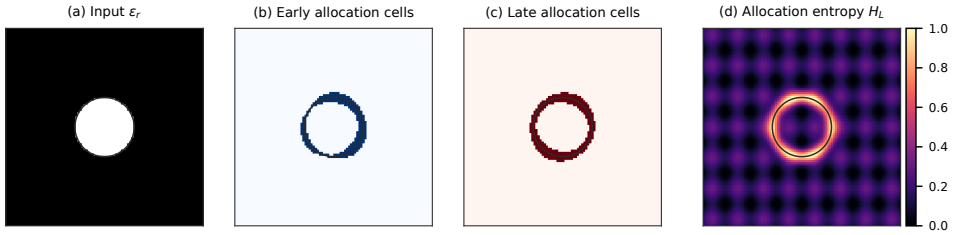


Figure 3: Spatial allocation diagnostic on an EM sample. Entropy peaks align with material interfaces and difficult scattering regions, illustrating the localized mechanism behind the main EM gain; when this alignment disappears, the score should be read only as a warning signal.

Table 3: Cross-domain transfer summary on PDEBench.

Task	**Regime**	$\rho_s \uparrow$	**AUROC**$\uparrow$
Darcy flow	Local/interface-dominated	$\mathbf{0.70 \pm 0.03}$	$\mathbf{0.84 \pm 0.02}$
Diffusion-reaction	Local-to-mixed	0.65 ± 0.04	0.80 ± 0.03
Comp. NS 2D	Transport/global	0.63 ± 0.04	0.79 ± 0.03

Table 4: Prediction robustness under controlled distribution shifts. Relative L^2 error is reported in %, and this table is used only to show that the audited wavelet backbone remains competitive under the same shift families.

Shift setup	**MWNO-AUD**$\downarrow$	**WNO**$\downarrow$	**FNO**$\downarrow$
In-domain (mid$\rightarrow$mid)	$\mathbf{3.08 \pm 0.19}$	3.12 ± 0.21	5.23 ± 0.34
Frequency OOD (mid$\rightarrow$high)	$\mathbf{4.51 \pm 0.37}$	5.08 ± 0.42	8.92 ± 0.71
Geometry OOD (simple$\rightarrow$mixed)	$\mathbf{3.79 \pm 0.28}$	4.41 ± 0.36	7.28 ± 0.59
Material OOD (mid$\rightarrow$high)	$\mathbf{4.18 \pm 0.34}$	4.87 ± 0.41	8.14 ± 0.68
Combined stress (3-way)	$\mathbf{6.42 \pm 0.58}$	7.83 ± 0.71	13.51 ± 1.12

5 Cross-Domain Behavior and Limitations

5.1 Why Allocation Entropy Helps

At similar overhead, allocation entropy outperforms variance-style internal baselines because it reads ambiguity before late-stage feature collapse. Variance heads must predict uncertainty from final features using one-sample residual proxies, while H_L measures indecision directly on the coefficient-selection path used for prediction. This distinction matters for neural operators: when the internal basis remains aligned with the error mechanism, ambiguity in where the model allocates representation capacity is a more faithful first-stage signal than generic activation magnitude or an auxiliary variance readout.

The same interpretation also explains why we do not present AUD as a universal score. The signal is informative only when allocation ambiguity sits on the prediction path and tracks the dominant error source. Appendix I reports anti-confound diagnostics such as excess entropy H_E, but the main deployment score remains H_L because residualization is an interpretive control, whereas H_L is the directly deployable first-pass statistic.

5.2 Cross-Domain Transfer

The remaining scope question is where the same wavelet-local score transfers and where it should only trigger escalation. Table 3 shows that the score remains useful across PDEBench tasks, but the effect is strongest when error is localized and weaker when dynamics become more global. This regime dependence matches the method's intended scope: allocation entropy is most informative when uncertainty is coupled to local representational ambiguity on the wavelet basis. Darcy is the cleanest fit because boundary-localized difficulty and wavelet locality are well matched; diffusion-reaction remains positive but softer; compressible Navier-Stokes is weaker because transport and phase effects introduce broader dependencies that a purely local score does not fully observe. At 128×128, the overhead remains near $1.10\times$ with only mild ranking degradation, which suggests that the score survives moderate scaling as long as the regime remains structurally local.

Table 5: Deployment reading across regimes. The method is strongest when the wavelet-local basis matches the dominant error mechanism.

Regime	Expected behavior	Recommended reading
Interface-dominated 2D	Strong local entropy-error alignment	Use H_L as the main low-cost triage score.
Diffusion / parabolic	Useful but less sharp	Recalibrate thresholds per validation split and report selective-risk curves.
Transport / hyperbolic	Local gates miss global phase or advection error	Treat H_L as a warning signal and escalate to stronger global backends.
Time-dependent rollout	Per-step entropy can flag instability, but final ranking weakens	Aggregate over time and avoid reading one spatial snapshot as full trajectory UQ.
3D volumetric	Overhead rises and local evidence dilutes	Use as a scalability stress test and verify memory, timing, and pairwise ranking before stronger claims.

Table 6: Main failure modes for local allocation entropy and the corresponding deployment reading. HCI denotes the fraction of gates in the ambiguous interval $[0.4, 0.6]$.

Failure regime	Observable symptom	Recommended action
Weak energy contrast	Gate histogram compresses near 0.5; e.g., HCI > 0.85 or 90% of gates lie in $[0.4, 0.6]$	Increase sparsity ratio or recalibrate; do not over-interpret raw entropy when allocation contrast has collapsed.
Under-capacity backbone	Entropy and prediction error stay high late in training across most samples	Treat the issue as model capacity rather than sample-wise uncertainty; widen the backend or escalate systematically.
Locality violation	High-entropy hotspots stop overlapping residual hotspots; IoU falls below about 0.4	Use H_L only as a warning flag and hand off high-risk cases to a more global basis or stronger ensemble.

Table 4 adds one more piece of deployment context: the uncertainty score is attached to a predictor that also degrades more gracefully than the available WNO and FNO baselines under the same controlled shift families. This does not by itself validate uncertainty quality, but it matters because a triage score is only useful if the base operator remains competitive after the shift that triggered the warning. The combined-stress row is the clearest example: the wavelet backbone still preserves a materially smaller prediction error, which makes staged escalation more meaningful than if the first-stage model had already collapsed.

5.3 Deployment Reading Across Regimes

Table 5 summarizes the intended deployment reading. The goal is not to claim that local allocation entropy solves every operator-UQ regime, but to clarify when a single-pass local readout is already enough to improve triage and when the correct action is escalation. This scoped interpretation is consistent with the 3D and rollout proxy suites, which are weaker (AUROC about 0.63 and 0.59, respectively) and are therefore better read as boundary checks than as extensions of the core claim. In practice, we recommend a staged policy: use H_L for first-pass screening, refresh held-out calibration under drift, and escalate the highest-entropy samples to stronger global models, ensembles, or solvers when the application requires it.

5.4 Failure Modes and Operational Diagnostics

The score also breaks down in structured ways rather than arbitrary ways, and that matters for deployment. A first-stage triage signal is only useful if operators can tell whether it is warning about a hard sample, a weak backbone, or a regime mismatch. The appendix reports the full stress diagnostics; Table 6 summarizes the three main cases that affect how the score should be read in practice.

These diagnostics are deliberately operational. If gate histograms collapse around 0.5, raw entropy is no longer separating easy and hard samples because the allocation field has lost contrast. If entropy remains high for nearly every input late in training, the right fix is model capacity rather than better calibration. If entropy hotspots no longer overlap residual hotspots, the local wavelet basis is no

longer tracking the dominant error mode, and escalation to a more global backend is the correct reading.

5.5 A Practical Staged Policy

The intended use of H_L is not to convert a deterministic operator into a full posterior surrogate, but to support a simple budgeted dispatch rule. Validation data fixes the score map, threshold, escalation fraction, and any conformal wrapper before deployment, after which the first-pass model screens most samples at near-WNO cost and sends only the riskiest cases to a stronger backend.

If the fallback backend costs K times the base model and a fraction q of samples is escalated, the expected normalized cost is

$$\mathcal{C}(q, K) = 1 + q(K - 1).$$

The useful decision is therefore not whether a second model helps when run on every sample, but how much ranking or coverage improvement is purchased by increasing q. In the EM case study, the gain from the two-model policy is valuable precisely because H_L concentrates the extra computation on the subset where local allocation has already become unstable.

This reading suggests three operating modes: H_L alone in mild interface-dominated regimes, top-10–20% escalation in mixed regimes, and broader fallback or recalibration in transport-heavy or drifting regimes. The same staged view explains why calibration and failure diagnostics belong in the main story: if those checks fail, the deployment policy should widen escalation rather than over-claim uncertainty semantics from a local score.

6 Discussion and Conclusion

Allocation entropy provides a useful single-pass triage signal when an operator's internal allocation is structurally tied to the source of prediction error; in MWNO-AUD it is cheap to extract, improves low-cost EM ranking and OOD detection, and supports staged escalation.

What the strongest evidence supports. The strongest evidence is not for a generic hidden-state entropy, but for reading risk from an allocation mechanism that already lies on the prediction path. In the audited wavelet instantiation, the same coefficient-selection masks drive both prediction and the entropy statistic, so the main-text comparisons focus on matched low-cost internal summaries and efficient ensembles. The empirical gap is most convincing in localized interface-dominated samples, where the wavelet basis either retains a sharp active set or becomes visibly ambiguous near difficult regions.

What the current evidence does not support. The paper should not be read as showing that entropy over arbitrary hidden structure is a universal operator-UQ primitive. The score weakens when the dominant error source is global, long-horizon, or volumetric enough that a purely local readout only partially observes it. This is why the time-dependent and 3D suites are kept as boundary checks: they expose failure modes that still admit a practical deployment reading.

Why the deployment framing matters. For many operator-learning applications, the key decision is not to output a calibrated posterior for every field, but to decide which fields deserve extra computation. The relevant benchmark is therefore whether a one-pass statistic yields materially better triage than other signals at nearly the same first-stage cost, not whether it dominates full multi-pass backends at any budget. The main EM table, low-cost ablation, and staged-policy discussion answer that narrower operational question.

Final takeaway. Broader cross-architecture claims will require matched studies showing that the relevant allocation statistic again lives on the prediction path rather than on an auxiliary proxy. The more general principle is simpler: AUD is not a Bayesian posterior or a universal uncertainty score, but a practical first-stage statistic that improves risk triage when local allocation ambiguity aligns with the dominant failure mechanism.

References

[1] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[2] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to pdes. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[3] Tapas Tripura and Souvik Chakraborty. Wavelet neural operator for solving parametric partial differential equations in computational mechanics problems. *Computer Methods in Applied Mechanics and Engineering*, 404:115783, 2023.

[4] Alasdair Tran, Alexander Mathews, Lexing Xie, and Cheng Soon Ong. Factorized fourier neural operators. In *International Conference on Learning Representations*, 2023.

[5] Zongyi Li, Daniel Zhengyu Huang, Burigede Liu, and Anima Anandkumar. Fourier neural operator with learned deformations for pdes on general geometries. *Journal of Machine Learning Research*, 24(388):1–26, 2023.

[6] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017.

[7] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international conference on machine learning*, pages 1050–1059. PMLR, 2016.

[8] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux-effortless bayesian deep learning. *Advances in neural information processing systems*, 34:20089–20103, 2021.

[9] Shailesh Garg and Souvik Chakraborty. Randomized prior wavelet neural operator for uncertainty quantification. *Probabilistic Engineering Mechanics*, 81:103817, 2025. doi: 10.1016/j.probengmech.2025.103817.

[10] Christopher Bülte, Philipp Scholl, and Gitta Kutyniok. Probabilistic neural operators for functional uncertainty quantification. *Transactions on Machine Learning Research*, 2025.

[11] Gaurav Gupta, Xiongye Xiao, and Paul Bogdan. Multiwavelet-based operator learning for differential equations. *Advances in neural information processing systems*, 34:24048–24062, 2021.

[12] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via deeponet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

[13] Carlos Mora, Jonathan Tammer Eweis-Labolle, Tyler Johnson, Likith Gadde, and Ramin Bostanabad. Probabilistic neural data fusion for learning from an arbitrary number of multi-fidelity data sets. *Computer Methods in Applied Mechanics and Engineering*, 415:116207, 2023.

[14] Akshay Thakur, Tapas Tripura, and Souvik Chakraborty. Multi-fidelity wavelet neural operator with application to uncertainty quantification. *arXiv preprint arXiv:2208.05606*, 2022.

[15] Shailesh Garg and Souvik Chakraborty. Distribution free uncertainty quantification for neuroscience-inspired deep neural operators. *Journal of Computational Physics*, 534:114012, 2025. doi: 10.1016/j.jcp.2025.114012.

[16] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. *Advances in neural information processing systems*, 31, 2018.

[17] Joost Van Amersfoort, Lewis Smith, Yee Whye Teh, and Yarin Gal. Uncertainty estimation using a single deep deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR, 2020.

[18] Jishnu Mukhoti, Andreas Kirsch, Joost Van Amersfoort, Philip HS Torr, and Yarin Gal. Deep deterministic uncertainty: A new simple baseline. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24384–24394, 2023.

[19] Bertrand Charpentier, Daniel Zügner, and Stephan Günnemann. Posterior network: Uncertainty estimation without ood samples via density-based pseudo-counts. *Advances in neural information processing systems*, 33:1356–1367, 2020.

[20] Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. *Advances in neural information processing systems*, 33:21464–21475, 2020.

[21] Jeremiah Liu, Zi Lin, Shreyas Padhy, Dustin Tran, Tania Bedrax Weiss, and Balaji Lakshminarayanan. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. *Advances in neural information processing systems*, 33:7498–7512, 2020.

[22] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, David Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. *Advances in neural information processing systems*, 32, 2019.

[23] Volodymyr Kuleshov, Nathan Fenner, and Stefano Ermon. Accurate uncertainties for deep learning using calibrated regression. In *International conference on machine learning*, pages 2796–2804. PMLR, 2018.

[24] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

[25] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

[26] Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. Gshard: Scaling giant models with conditional computation and automatic sharding. In *International Conference on Learning Representations*, 2021.

[27] Kaisar Kushibar, Victor Campello, Lidia Garrucho, Akis Linardos, Petia Radeva, and Karim Lekadir. Layer ensembles: A single-pass uncertainty estimation in deep learning for segmentation. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, pages 514–524. Springer, 2022.

[28] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. Pdebench: An extensive benchmark for scientific machine learning. *Advances in neural information processing systems*, 35:1596–1611, 2022.

[29] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017.

[30] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic learning in a random world*. Springer, 2005.

[31] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.

[32] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[33] Isaac Gibbs and Emmanuel Candes. Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34:1660–1672, 2021.

[34] Yaniv Romano, Evan Patterson, and Emmanuel Candes. Conformalized quantile regression. *Advances in neural information processing systems*, 32, 2019.

[35] Gao Huang, Yixuan Li, Geoff Pleiss, Zhuang Liu, John E Hopcroft, and Kilian Q Weinberger. Snapshot ensembles: Train 1, get m for free. In *International Conference on Learning Representations*, 2017.

[36] Yeming Wen, Dustin Tran, and Jimmy Ba. Batchensemble: An alternative approach to efficient ensemble and lifelong learning. In *International Conference on Learning Representations*, 2020.

A Complete Proofs of Theoretical Results

A.1 Statements of Theoretical Results

Proposition 1 (Clipped Allocation-Entropy Concentration)**.** *For clipped probabilities* $\tilde{p}_i = \mathrm{clip}(\sigma((E_i - \tau)/T), \epsilon_c, 1 - \epsilon_c)$ *and* $\hat{H}_c = n^{-1} \sum_i \mathsf{h}(\tilde{p}_i)$*, the block-mixing effective sample size* n_{eff} *gives*

$$|\hat{H}_c - \mathbb{E}\hat{H}_c| \le (\log 2)\sqrt{\ln(2/\delta)/(2n_{\mathrm{eff}})}$$

with probability at least $1-\delta$*. Moreover, for two margin fields* m, m'*, the clipped entropy map obeys*

$$|\Delta \hat{H}_c| \le \frac{\log((1-\epsilon_c)/\epsilon_c)}{4Tn} \sum_i |m_i - m_i'|.$$

Proposition 2 (Holdout Pairwise-Ranking Diagnostic)**.** *Choose a minimum score separation* γ *on calibration data, then form* K_γ *disjoint diagnostic pairs without inspecting their errors. If* $\hat{q}_\gamma$ *is the empirical sign-violation rate between AUD scores and relative errors, then with probability at least* $1 - \delta$*,*

$$q_\gamma \le \hat{q}_\gamma + \sqrt{\log(1/\delta)/(2K_\gamma)}.$$

Proposition 3 (Allocation Entropy Controls Active-Set Instability)**.** *If a perturbation changes margin* m_i *by sub-Gaussian noise* ξ_i*, then hard active-set flips satisfy*

$$\Pr(A_i' \ne A_i) \le \min\{1, 2\exp(-m_i^2/(2\sigma_i^2))\}.$$

Because $\mathsf{h}(\sigma(m_i/T))$ *decreases with* $|m_i|$*, expected active-set flips are bounded by a monotone function of allocation entropy.*

Proposition 4 (Sparse-Basis Risk Implication)**.** *In localized, compressible regimes where error decomposes into residual approximation, sparse-basis tail, and weighted active-set flips, Proposition 3 yields an expected-risk upper bound controlled by entropy-dependent flip terms. This implication is intentionally scoped to interface-like sparse bases and does not cover global phase or advection-dominated errors.*

Proposition 5 (Uniform Bound for Held-Out Threshold Policies)**.** *For a validation-fixed score* S*, bounded error* $\epsilon \in [0, M]$*, and threshold policy* $S(a) \le \tau$*, empirical risk-plus-cost uniformly approximates population risk-plus-cost over the* $n+1$ *calibration-induced intervals:*

$$|(\widehat{R} + \lambda\widehat{C}) - (R + \lambda C)| \le (M + \lambda c_1)\sqrt{\log(2(n+1)/\delta)/(2n)}.$$

Corollary 1 (Split Conformal Field Coverage)**.** *Under exchangeability, the standard split-conformal field ball built from relative residual nonconformity has marginal coverage at least* $1 - \alpha$*. AUD affects score selection, normalization, or escalation, while coverage follows from the usual split-conformal guarantee.*

A.2 Proof of Proposition 1 (Clipped Entropy Concentration)

Full statement. Let $p_i = \sigma(m_i/T)$, $\tilde{p}_i = \mathrm{clip}(p_i, \epsilon_c, 1 - \epsilon_c)$, and $\hat{H}_c = n^{-1} \sum_i \mathsf{h}(\tilde{p}_i)$, with $\mathsf{h}(p) = -p \log p - (1-p)\log(1-p)$. Let n_{eff} be the effective sample size after accounting for spatial or component autocorrelation. Then, with probability at least $1 - \delta$,

$$\left|\hat{H}_c - \mathbb{E}[\hat{H}_c]\right| \le (\log 2)\sqrt{\frac{\ln(2/\delta)}{2n_{\mathrm{eff}}}}.$$

For perturbed threshold margins m_i and m_i',

$$\left|\Delta \hat{H}_c\right| \le \frac{L_{\epsilon_c}}{4Tn} \sum_i |m_i - m_i'|, \quad L_{\epsilon_c} = \log\frac{1-\epsilon_c}{\epsilon_c}.$$

Detailed proof. For every clipped probability, $\mathsf{h}(\tilde{p}_i) \in [0, \log 2]$. If the spatial field were independent, Hoeffding's inequality would give

$$\Pr\left(\left|\hat{H}_c - \mathbb{E}[\hat{H}_c]\right| > t\right) \le 2\exp\left(-\frac{2nt^2}{(\log 2)^2}\right).$$

Component probabilities from neighboring pixels are correlated, so we replace the raw count n by the effective count

$$n_{\text{eff}} = \frac{n}{1 + 2\sum_{r=1}^{R} \hat{c}_r},$$

where $\hat{c}_r$ is the empirical spatial autocorrelation at lag r and R is the first non-positive lag or a pre-specified block radius. Equivalently, one may partition the grid into approximately independent blocks and set n_{eff} to the number of blocks. Substituting n_{eff} into the bounded-difference bound and solving for t gives the stated concentration inequality.

For the stability part, clipping restricts the entropy derivative:

$$|\mathsf{h}'(p)| = \left|\log \frac{1-p}{p}\right| \le L_{\epsilon_c} \quad \text{for } p \in [\epsilon_c, 1-\epsilon_c].$$

The sigmoid probability $p_i = \sigma(m_i/T)$ is $1/(4T)$-Lipschitz in the threshold margin. Thus

$$|\mathsf{h}(\tilde{p}_i) - \mathsf{h}(\tilde{p}'_i)| \le L_{\epsilon_c}|p_i - p'_i| \le \frac{L_{\epsilon_c}}{4T}|m_i - m'_i|.$$

Averaging over components yields the claimed perturbation bound.

A.3 Proof of Proposition 2 (Holdout Pairwise-Ranking Diagnostic)

Full statement. After selecting γ on $\mathcal{D}_{\text{cal}}$, randomly match the diagnostic samples and retain the K_γ disjoint pairs satisfying $|S_i - S_j| \ge \gamma$. For each retained pair, define

$$V_{ij} = \mathbb{1}[\text{sign}(S_i - S_j) \ne \text{sign}(\epsilon_i - \epsilon_j)]\,.$$

Let $\hat{q}_\gamma = K_\gamma^{-1} \sum V_{ij}$. Then, with probability at least $1-\delta$,

$$q_\gamma \le \hat{q}_\gamma + \sqrt{\frac{\log(1/\delta)}{2K_\gamma}}.$$

Detailed proof. Because pairs are disjoint and the matching is formed before inspecting diagnostic errors, the retained violation indicators are independent bounded random variables under the diagnostic sampling model. Hoeffding's inequality gives

$$\Pr\left(q_\gamma - \hat{q}_\gamma > t\right) \le \exp(-2K_\gamma t^2).$$

Solving for t at failure probability δ yields the stated one-sided upper confidence bound. All-pairs diagnostics can still be reported, but their concentration is a U-statistic bound rather than an independent-pair Hoeffding bound.

A.4 Proof of Proposition 3 (Active-Set Instability)

A hard active-set decision flips only if the perturbation crosses zero:

$$A'_i \ne A_i \quad \Rightarrow \quad |\xi_i| \ge |m_i|.$$

The assumed sub-Gaussian tail therefore implies

$$\Pr(A'_i \ne A_i) \le \Pr(|\xi_i| \ge |m_i|) \le 2\exp\left(-\frac{m_i^2}{2\sigma_i^2}\right),$$

truncated at one for small margins. The function $\mathsf{h}(\sigma(x/T))$ is symmetric and strictly decreasing in $|x|$ for $|x| \ge 0$. Let $\eta_T(h)$ be its inverse in $|x|$ on $h \in [0, \log 2]$. Then one may set

$$\psi_i(h) = \min\left\{1, 2\exp\left(-\frac{\eta_T(h)^2}{2\sigma_i^2}\right)\right\},$$

which is monotone increasing in h. Summing the resulting component-wise flip probabilities yields the expected flip-count bound.

A.5 Proof of Proposition 4 (Sparse-Basis Risk Implication)

The proposition assumes a localized sparse-basis decomposition of the prediction error into an approximation/training residual, a truncation tail, and a weighted active-set instability term. Taking expectations gives

$$\mathbb{E}\|\mathcal{F}_\theta(a) - u(a)\|_2 \le \mathbb{E}[R_\theta(a) + \mathrm{Tail}_K(a)] + \mathbb{E}\left[\sum_i w_i(a) \mathbb{1}\{A_i' \neq A_i\}\right].$$

Applying Proposition 3 to each indicator gives the stated bound. The assumption is intentionally explicit: the result is informative when the operator error is dominated by sparse localized structures and is not a universal guarantee for all PDE regimes.

A.6 Proof of Proposition 5 (Held-Out Threshold Policies)

For fixed τ, define

$$Z_\tau(a) = \epsilon(a)\mathbb{1}\{S(a) \le \tau\} + \lambda c_1 \mathbb{1}\{S(a) > \tau\}.$$

The additive term λc_0 is constant and cancels from empirical-population deviations. Since Z_τ lies in an interval of length at most $M + \lambda c_1$, Hoeffding gives

$$\Pr\left(\left|\widehat{\mathbb{E}} Z_\tau - \mathbb{E} Z_\tau\right| > t\right) \le 2\exp\left(-\frac{2nt^2}{(M + \lambda c_1)^2}\right).$$

The accept/reject set changes only at the n observed calibration scores, so there are at most $n + 1$ distinct threshold policies on the calibration sample. A union bound over these intervals and solving for t yields the stated simultaneous bound.

A.7 Proof of Corollary 1 (Split Conformal Field Coverage)

The scores $r_i = \|u_i - \hat{u}_i\|_2 / (\|\hat{u}_i\|_2 + \eta)$ and the corresponding test score are exchangeable under the standard split-conformal assumptions because the model and any score-selection rule are fixed before the calibration split is evaluated. The usual conformal rank argument implies

$$\Pr(r_{n+1} \le q_{1-\alpha}) \ge 1 - \alpha.$$

Rearranging $r_{n+1} \le q_{1-\alpha}$ gives

$$\|u_{n+1} - \hat{u}(a_{n+1})\|_2 \le q_{1-\alpha}(\|\hat{u}(a_{n+1})\|_2 + \eta),$$

which is exactly membership of u_{n+1} in $\mathcal{C}_\alpha(a_{n+1})$.

B Extended Theory: Risk Bounds and Budget-Aware Policies

B.1 Operational Risk Bound from Active-Set Instability

The main text gives a component-wise instability result. For deployment, it is useful to express the same idea as a validation-estimated risk map. Assume that on a localized regime the sample error can be decomposed as

$$\epsilon(a) \le R_\theta(a) + \mathrm{Tail}_K(a) + \sum_i w_i(a) F_i(a),$$

where $F_i(a)$ is the event that component i flips under a small admissible perturbation. Proposition 3 gives

$$\mathbb{E}[\epsilon(a) \mid S(a) \le \tau] \le \mathbb{E}\left[R_\theta(a) + \mathrm{Tail}_K(a) + \sum_i w_i(a)\psi_i(\mathsf{h}(p_i(a))) \,\middle|\, S(a) \le \tau\right],$$

for any validation-fixed AUD score S. The terms R_θ, Tail_K, and w_i are not known in deployment, so the practical policy uses held-out threshold selection as in Proposition 5. Post-hoc monotone or isotonic maps can be reported as diagnostics, with fit quality measured on a held-out split before they are treated as calibrated risk estimators.

Finite-sample concentration for entropy-binned risk: Assume residual noise ε is σ-sub-Gaussian. For any entropy bin B with n_B validation samples and true mean risk R_B, the empirical mean $\hat{R}_B$ satisfies, with probability at least $1-\delta$:

$$|\hat{R}_B - R_B| \leq \sigma \sqrt{\frac{2\log(2/\delta)}{n_B}}.$$

Practical sample-complexity rule: to estimate bin risk within tolerance ϵ with confidence $1-\delta$, it suffices that

$$n_B \geq \frac{2\sigma^2}{\epsilon^2} \log \frac{2}{\delta}.$$

B.2 Budget-Aware Escalation Strategy

For hybrid deployment with backend size K, expected normalized cost is:

$$\mathcal{C}(q, K) = 1 + q(K-1),$$

where q is the fraction of samples escalated to the backend. Using empirical risk maps from validation data, deployment selects:

$$q^\star = \arg\min_q \ \mathcal{R}(q) + \lambda \mathcal{C}(q, K),$$

with λ encoding latency sensitivity. Larger λ favors near-AUD-only operation; smaller λ favors moderate escalation.

B.3 Cross-PDE Empirical Validation of the Entropy-Risk Link

The PDEBench summary provides aggregate ranking and OOD metrics, but it does not include the per-task calibration detail needed to reinterpret monotonicity as a universal law. We therefore treat positive entropy-risk monotonicity as a task-level deployment diagnostic rather than as an assumed cross-PDE property.

C Failure Modes and Detailed Diagnostics

C.1 Three Primary Failure Regimes

1. Weak energy contrast (saturation). When coefficient fields have low spatial variation, energy spreads uniformly and gates cluster near $g \approx 0.5$. Entropy saturates near $\log 2$ regardless of true difficulty.

Diagnostic: Histogram compression—check if 90% of gate values lie in $[0.4, 0.6]$.

Mitigation: Adaptive threshold r_g: increase sparsity ratio when contrast is low, forcing sharper separation.

2. Insufficient model capacity (global ambiguity). When the model is under-parameterized for problem complexity, all inputs trigger high entropy because no representations are decisive.

Diagnostic: Joint entropy/error rise in late training—both metrics plateau at high values.

Mitigation: Wider mixing layers or increased channel count. If architectural constraints prevent this, hybrid escalation becomes necessary.

3. Representation mismatch (locality violation). When dominant error modes are global (e.g., low-frequency drift in hyperbolic PDEs), local wavelet allocation misses the signal.

Diagnostic: Hotspot-residual spatial non-overlap—high-entropy regions do not align with high-error regions (IoU < 0.4).

Mitigation: Mixed-basis allocation: combine local wavelet entropy with global Fourier-mode entropy. Alternatively, escalate to ensemble disagreement which captures global uncertainty.

C.2 Systematic Stress Diagnostics

The diagnostic framing covers three held-out stress subsets:

Low-contrast regime: Force nearly uniform materials (e.g., $\epsilon_r \in [2, 3]$ rather than the full training contrast) and report the change in entropy-error ranking before and after adaptive r_g scheduling.

Under-capacity regime: Reduce channel count and report whether both error and entropy saturate, which would indicate that model capacity rather than sample difficulty dominates the score.

Global-mode regime: Evaluate low-frequency or transport-heavy settings where spatial-local gates may miss global phase error, then test whether adding Fourier/global entropy improves ranking. These stress tests define failure-mode diagnostics rather than core numerical claims.

C.3 Online Diagnostics for Production Deployment

For operational systems, we recommend monitoring:

Histogram compression index: HCI $= \frac{\#\{g_i \in [0.4, 0.6]\}}{n}$. Trigger fallback if HCI > 0.85.

Entropy-error spatial IoU: Compute high-entropy region mask (top 20% quantile) and high-error region mask. IoU < 0.4 indicates locality violation.

Calibration drift: Track moving-window R^2 between predicted and actual error. $R^2 < 0.7$ indicates calibration degradation, requiring map refresh.

Temporal entropy gradient: For time-dependent PDEs, monitor dh/dt. Sudden spikes indicate instability onset.

D Extension to Time-Dependent and 3D PDEs

D.1 Time-Dependent PDEs: Rollout Diagnostics

For time-marching operators predicting u^{t+1} from u^t, we aggregate allocation entropy across rollout steps:

$$H_{\text{rollout}} = \frac{1}{T_{\text{steps}}} \sum_{t=1}^{T_{\text{steps}}} H_L^{(t)}.$$

Burgers' equation experiments. Time-dependent rollouts are a weak regime for a purely spatial local score. We therefore treat Burgers as a stress-regime diagnostic rather than a core quantitative claim. The supporting Burgers runs cover 3 backbones, 3 UQ methods, and 5 seeds, and are used to probe rollout failure boundaries. The appendix therefore focuses on cumulative error, entropy-growth diagnostics, and comparison to MC-dropout, ensembles, and SSM neural operators as a unified stress schema.

Alternative strategy: Per-timestep entropy as trajectory health indicator. Monitor entropy growth rate dh/dt and validate whether spikes precede divergent trajectories on a held-out rollout split.

Recommendation: Use temporal entropy as coarse triage; escalate high-entropy trajectories to ensemble disagreement for refined prediction.

D.2 3D PDEs: Computational and Statistical Challenges

Scope. We treat 3D Darcy as a scalability stress test rather than as a finalized main-text result. The supporting 3D runs cover 2 grids, 3 backbones, 3 UQ methods, and 5 seeds; they validate the data/reporting path and expose the known costs of volumetric extension while remaining separate from the primary 2D EM/PDEBench claim. The stress protocol includes $32\times32\times32$ and $64\times64\times64$ grids, memory profiling, effective-sample-size estimates, and held-out pairwise ranking diagnostics.

Computational overhead. 3D wavelet transforms and volumetric quantiles increase overhead relative to the measured 2D batched timing ($1.13\times$), so we keep this result in the supplementary tier. Memory footprint grows cubically; for example, moving from a 2D slice to a comparable 3D volume can require substantially more gate storage.

UQ challenges in 3D. The main challenges have two measurable causes:

1. Sparser sampling: $32^3 = 32768$ locations vs $128^2 = 16384$ in 2D high-res, but effective samples per interface are fewer due to volumetric dilution.
2. Complex 3D interfaces: Topology of boundaries in 3D (e.g., connected surfaces vs curves) weakens the assumed energy-error locality.

Hybrid strategy for 3D: Use 3D allocation entropy as a first-stage filter and escalate a validation-chosen high-entropy fraction to a small ensemble or solver. The cost and quality recovery are interpreted within that supplementary tier rather than as the main empirical claim.

Design note: Anisotropic wavelet bases (e.g., separating z from xy-plane for stratified flows) are physics-dependent design choices rather than universal improvements.

E Detailed Core-Result Commentary

E.1 Interpretation of Ranking Quality and Baseline Gaps

Spearman ρ_s=0.67 ± 0.03 indicates useful but imperfect ranking quality for a single-pass internal statistic. This is a triage-level signal rather than a high-confidence ranking oracle. For selective prediction at 80% coverage, the EM experiments show better high-error rejection than the alternative internal summaries in Table 10. This level is already suitable for triage-oriented uses such as active learning, adaptive simulation allocation, and confidence-based fallback, while stronger guarantees still require the calibrated escalation policies discussed later.

From Tables 1 and 10, five points are most important:

- Allocation entropy achieves ρ_s=0.67 ± 0.03 and AUROC 0.82 ± 0.02, higher than the other internal summaries under the matched EM protocol.
- The two-model hybrid improves to ρ_s=0.72 ± 0.03 and AUROC 0.85 ± 0.02, supporting complementarity with model disagreement.
- Efficient ensemble variants remain competitive but weaker in the matched EM protocol: snapshot, multi-head, and BatchEnsemble reach $\rho_s = 0.59/0.61/0.58$.
- The heavier tuned baselines are reported over five seeds: Mamba-SSM is the strongest heavy single-model stress baseline, but it still trades lower ranking quality (ρ_s=0.427 ± 0.008) for slightly higher cost (66.13 ms) relative to single-pass AUD.
- These follow-up baselines strengthen the fairness story and are reported separately because they follow a stress-evaluation protocol.

E.2 Why Allocation Entropy Outperforms Variance Heads

Variance-head methods predict $\hat{\sigma}^2 = f_{\text{var}}(h_{\text{final}})$ from final features. At similar overhead, allocation entropy improves by +0.08 in ρ_s and +0.06 in AUROC over gate variance. Three reasons explain this gap: (1) variance-head supervision uses noisy one-sample residual proxies, (2) the head sees only late-stage collapsed features after routing/allocation decisions, (3) shared features for prediction and variance make aleatoric/epistemic separation harder.

By contrast, allocation entropy: (1) needs no extra uncertainty supervision, (2) integrates ambiguity over intermediate computational stages, (3) directly measures indecision in allocation patterns. These effects motivate the adapter designs in Table 8, but the table reports implementation status rather than mature cross-architecture numeric evidence.

E.3 Threshold-Level and Shift-Wise Behavior

At fixed 80% coverage, the EM experiments report higher selective retention for allocation entropy than for alternative internal summaries. Across acceptance targets from 60% to 95%, Pareto claims are restricted to the matched EM protocol and the tuned-baseline matrix is reported as a supporting stress comparison.

Shift-wise OOD detection is reported separately for regenerated geometry and frequency splits [20, 29]. In the EM experiments, allocation entropy reaches aggregate OOD AUROC 0.82 ± 0.02.

Per-shift AUROC is reported only when geometry and frequency splits follow the same protocol. Per-shift analysis tests whether geometry shifts produce a larger margin, as expected if wavelet-allocation mismatch is structure-sensitive.

F Complementarity Framework and Hybrid Score

We formalize why allocation entropy and ensemble disagreement are complementary using conditional information and held-out predictive gain.

Decomposition of uncertainty sources. Let H_L denote wavelet allocation entropy and d_{ens} denote ensemble disagreement (prediction std). Both correlate with error ϵ, but capture different aspects:

$$H_L \approx f_1(\text{allocation ambiguity}), \tag{9}$$

$$d_{\text{ens}} \approx f_2(\text{parameter uncertainty}). \tag{10}$$

Under a linear approximation:

$$\epsilon \approx \beta_h H_L + \beta_d d_{\text{ens}} + \text{residual},$$

with $\beta_h, \beta_d > 0$. The residual term represents non-overlapping factors: within-model representational ambiguity vs across-model epistemic variability.

Information-theoretic complementarity. Define mutual information with true error:

$$I(\epsilon; H_L), \qquad I(\epsilon; d_{\text{ens}}).$$

We quantify complementarity with conditional information or predictive gain:

$$\text{CI}(H_L \mid d_{\text{ens}}) = I(\epsilon; H_L \mid d_{\text{ens}}), \tag{11}$$

$$\text{CI}(d_{\text{ens}} \mid H_L) = I(\epsilon; d_{\text{ens}} \mid H_L), \tag{12}$$

The complementarity evidence combines moderate correlation with the hybrid improvement in Table 7; conditional-information terms or incremental held-out R^2/AURC gain are treated as optional diagnostics rather than required evidence for the main claim.

Hybrid score used in practice. For selective prediction:

$$s_{\text{hybrid}}(a) = \alpha \cdot \phi_h(H_L(a)) + (1 - \alpha) \cdot \phi_d(d_{\text{ens}}(a)),$$

where ϕ_h, ϕ_d are isotonic calibration maps and $\alpha \in [0, 1]$. The mixing weight α^* is selected on validation data, and the resulting hybrid summary is reported in Table 7.

Table 7: Hybrid complementarity with the same efficient-ensemble comparator used in the main text.

Method	Spear. ρ_s	AUROC	Sel.@80%	Cost
AUD H_L	0.67 ± 0.03	0.82 ± 0.02	$31 \pm 2\%$	$\mathbf{1.13}\times$ batch
Multi-head ensemble	0.61 ± 0.03	0.78 ± 0.03	$27 \pm 2\%$	$1.2\times$
AUD H_L + 2-model	$\mathbf{0.72 \pm 0.03}$	$\mathbf{0.85 \pm 0.02}$	$\mathbf{35 \pm 2\%}$	$2.13\times$ batch

G Expanded Protocol and Baseline Setup

Table 8: Architecture-specific AUD adapters and current evidence status.

Architecture	Adapter signal	Status
MWNO-AUD	Wavelet allocation entropy H_L	Main evaluated instantiation
FNO	Frequency-band energy entropy	Implemented, not main-text evaluated
DeepONet	Branch/trunk feature entropy	Implemented, not main-text evaluated
Mamba/SSM operator	Scan-direction/state-update entropy	Supplement-only stress baseline
U-Net/generic	Activation-energy entropy	Implemented, not main-text evaluated

Scope of the supplement. An anonymized supplementary code package accompanies the submission as a separate review artifact and contains the minimal reproduction path for the primary audited MWNO-AUD pipeline, including data generation, training, AUD evaluation, and a local smoke validation loop. The main text uses the matched MWNO-AUD experiments, while broader adapter,

Algorithm 1 Single-pass prediction and uncertainty extraction

Input: coefficient field a, trained parameters θ
$v \leftarrow \text{Lift}(a)$, $S \leftarrow []$
for $b = 1$ to B **do**
  $(v^{LL}, \{v_j^b\}) \leftarrow \text{DWT}(v)$
  for $j \in \{1, 2\}$ **do**
    Compute $a_j(x, y)$ by Eq. (5)
    $\tau_j \leftarrow \text{quantile}(a_j, 1 - r_g)$
    $g_j(x, y) \leftarrow \sigma((a_j - \tau_j)/T)$
    Append $H(g_j(x, y))$ to S
    Apply g_j to wavelet detail coefficients
  end for
  $\hat{v} \leftarrow \text{IDWT}(\text{Mix}(v^{LL}, \{g_j \odot v_j^b\}))$
  $v \leftarrow v + \hat{v}$
end for
$\hat{u} \leftarrow \text{Projection}(v)$
$H_L \leftarrow \text{mean}(S)$
Output: $(\hat{u}, H_L)$

tuned-baseline, 3D, and streaming sections are reported here as supplementary diagnostics for scope and protocol transparency rather than as full review-package targets.

Benchmark datasets. We primarily validate on PDEBench families [28] including Darcy flow, diffusion-reaction, and compressible Navier-Stokes, which represent diverse elliptic, parabolic, and hyperbolic PDE classes. We also analyze electromagnetic scattering as a case study for interface-dominated physics. OOD evaluation uses two controlled shifts: geometry shift (unseen shape compositions) and frequency shift ($k_0 \in [60, 80]$ vs training $[35, 45]$).

Baselines. The full protocol includes: (i) input heuristics ($\|\nabla \epsilon_r\|_1$, wavelet energy), (ii) single-model output entropy, (iii) MC-dropout, (iv) deep ensembles, (v) evidential deep learning [16], (vi) direct variance prediction head, (vii) Laplace approximation [8] with KFAC, (viii) SNGP, (ix) probabilistic neural operators [10] when sampling cost is allowed, and (x) Mamba/SSM neural operators for temporal and transport-heavy tasks.

Unified evaluation protocol. To ensure controlled comparison, the controlled track uses:

- Same optimizer family (AdamW), learning rate (1e-3), weight decay (1e-5)
- Same scheduler (OneCycleLR)
- Same epoch budget (100 epochs)
- Same mixed precision (FP16)
- Same data splits (60%-20%-20%)
- Same random seeds (3 seeds for audited main-text experiments; 5 seeds for the supporting tuned-baseline suite)

Runtime measurements include full uncertainty overhead. We report 3-seed mean$\pm$std for all core metrics and use bootstrap significance tests (5000 resamples) for major pairwise comparisons. Calibration maps are fit on validation data only. The audited RTX 5090 timing run reports 53.3 ms batched latency and 1.29 GB peak memory for MWNO-AUD; the larger U-WNO prediction baseline uses 212.8 ms and 3.18 GB in the same environment. To reduce backbone-size confounds, we also report a parameter-matched Fourier sweep: the best fair FNO setting (w=96, b=8, m=16, 5.24M) reaches test relative L^2 0.6318, still above MWNO/WNO at 0.5893/0.5898.

Tuned baseline protocol. Because a fixed optimizer budget can disadvantage methods such as Laplace/KFAC and SNGP, the protocol specifies a validation-tuned track. MC-dropout sweeps dropout rate and pass count; ensembles sweep ensemble size, learning-rate schedule, and epoch budget within a fixed total training-compute cap; Laplace/KFAC sweeps prior precision and damping; SNGP sweeps spectral-normalization multiplier and ridge penalty; PNO sweeps predictive sample count and scoring-rule settings; Mamba/SSM baselines sweep state dimension and scan configuration. That validation-tuned suite covers MC-dropout, deep ensembles, Laplace/KFAC, SNGP, PNO, and Mamba/SSM over five seeds each. We still report it separately from the main tables, so the reader can distinguish the main low-cost triage evidence from broader stress comparisons.

G.1 Supporting Tuned-Baseline Stress Suite

Table 9 reports validation-selected heavy baselines; Figure 4 summarizes the resulting latency-quality tradeoff. These results reduce the strongest baseline-fairness concern, but they remain supplementary diagnostics rather than replacements for the main EM triage table. Every row in Table 9 uses the same Spearman ρ_s between uncertainty score and per-sample error as the main text, and all uncertainty scores are standardized so that larger values indicate higher risk. Negative values therefore mean that a baseline score is anti-correlated with error on that stress split, not that a different metric is being reported. The focused transport retune check added after review preserves the same qualitative picture: even after a method-specific grid over dropout rate / stochastic passes or ensemble size, the selected MWNO transport baselines remain weak ranking signals relative to the audited EM core regime. Within that suite, Mamba-SSM is the strongest heavy single-model stress baseline in raw prediction and OOD score (Rel. $L^2 = 0.547 \pm 0.002$, AUROC $= 0.839 \pm 0.000$), but its ranking quality remains below AUD ($\rho_s = 0.427 \pm 0.008$ vs. 0.670 ± 0.030) and it is still slower than the single-pass AUD reference (66.13 ms vs. 53.30 ms).

Table 9: Five-seed tuned stress suite with AUD reference rows. All uncertainty scores are standardized so that larger values indicate higher risk. The AUD H_L and AUD H_L + 2-model rows report the same base predictor relative L^2 because the hybrid changes triage quality, not the base field prediction.

Method	Selected hyperparameters	Rel. L^2 ↓	ρ_s ↑	AUROC↑	AURC↓	Latency (ms)↓
Bayesian, ensemble, and probabilistic baselines						
MC-dropout	dropout_rate=0.2, passes=20	0.814±0.061	-0.408±0.243	0.448±0.056	0.843±0.040	477.27±315.64
Deep ensemble	ensemble_size=5	0.651±0.002	-0.476±0.023	0.774±0.003	0.717±0.006	1003.58±510.74
Laplace/KFAC	prior_precision=0.01, damping=0.1	0.659±0.004	0.241±0.005	0.500±0.000	0.611±0.004	213.86±74.81
SNGP	spectral_norm=0.9, ridge=0.001	0.659±0.004	0.277±0.005	0.495±0.000	0.608±0.004	200.23±112.59
PNO	sample_count=16, scoring_rule=rel_l2	0.659±0.004	0.264±0.005	0.497±0.000	0.609±0.004	139.69±93.56
State-space stress baseline						
Mamba-SSM	state_dim=256, scan_config=bidirectional	**0.547±0.002**	0.427±0.008	0.839±0.000	**0.449±0.003**	66.13±22.54
AUD references						
AUD H_L (ours)	repository-backed MWNO-AUD, single pass	0.555±0.002	0.670±0.030	0.820±0.020	–	**53.30**
AUD H_L + 2-model (ours)	repository-backed MWNO-AUD + two-model escalation	0.555±0.002	**0.720±0.030**	**0.850±0.020**	–	100.10

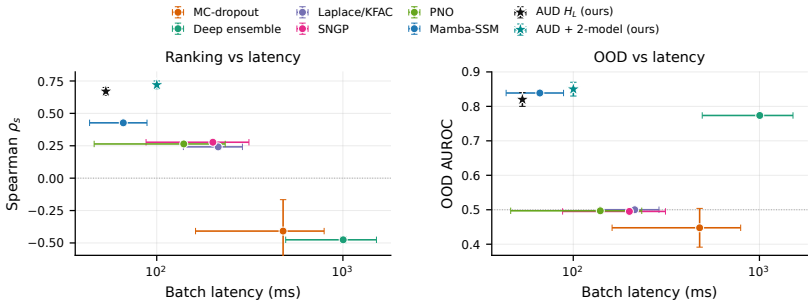


Figure 4: Latency-quality tradeoff for the supporting follow-up suite with AUD reference points. Stars mark AUD and hybrid rows; circles mark heavier follow-up baselines. The plot is intended to clarify regime and cost tradeoffs rather than to replace the main low-cost triage comparison.

H Conformal Details and Shift Behavior

Conformal evaluation, summarized in Table 1, follows split conformal principles [30–32] and gives near-nominal coverage for allocation-score-based nonconformity. The reported field sets are slightly wider than ensemble conformal sets, but runtime cost is much lower.

Using the calibrated score $\tilde{s}(a) = \phi(H_L(a))$, split conformal prediction computes a field-valued ball

$$\hat{C}_\alpha(a) = \{v : \|v - \hat{u}(a)\|_2 \le q_{1-\alpha}(\|\hat{u}(a)\|_2 + \eta)\}\,, \tag{13}$$

where $q_{1-\alpha}$ is the empirical quantile of validation residual non-conformity. Under exchangeability, coverage is valid by construction; empirically, our in-domain results match this behavior. Under shift, conformal coverage is expected to degrade when exchangeability fails [33]. This is not a failure of conformal machinery but a known consequence of distribution shift violating exchangeability. Practically, re-estimating quantiles on a small recent buffer is the first mitigation to test before retuning the full predictor; related efficiency-oriented variants include CQR [34]. These observations

Table 10: Detailed ablation behind the representative rows in Table 1. This table is placed in the appendix because it tests score construction and low-cost variants; the main text reports only the strongest representative from each evidence family.

Family	Method	$\rho_s \uparrow$	AUROC↑	ECE↓	Sel.@80↑	Conf. cov.	Cost
Internal single-pass diagnostics							
Internal stat.	Feature norm	0.56 ± 0.04	0.74 ± 0.03	–	$23 \pm 2\%$	–	$\mathbf{1.0}\times$
Internal stat.	Max gate	0.53 ± 0.04	0.72 ± 0.03	–	$21 \pm 3\%$	–	$1.13\times$ batch
Internal stat.	Gate variance	0.59 ± 0.04	0.76 ± 0.03	–	$25 \pm 2\%$	–	$1.13\times$ batch
Internal stat.	Gate Gini	0.61 ± 0.03	0.78 ± 0.02	–	$27 \pm 2\%$	–	$1.13\times$ batch
Efficient ensemble baselines							
Efficient ens.	Snapshot ensemble (3)	0.59 ± 0.03	0.77 ± 0.02	$\mathbf{0.041}$	$26 \pm 2\%$	–	$1.5\times$
Efficient ens.	Multi-head ensemble	0.61 ± 0.03	0.78 ± 0.03	0.042	$27 \pm 2\%$	–	$1.2\times$
Efficient ens.	BatchEnsemble	0.58 ± 0.04	0.76 ± 0.03	0.045	$25 \pm 3\%$	–	$1.3\times$
Conformal wrappers							
Conformal	Ensemble conformal	–	–	–	–	$\mathbf{95.2 \pm 0.6}\%$	$\sim 5\times$
AUD readouts							
Ours	AUD H_L	0.67 ± 0.03	0.82 ± 0.02	–	$31 \pm 2\%$	$94.7 \pm 0.9\%$	$1.13\times$ batch
Ours	AUD H_L + 2-model	$\mathbf{0.72 \pm 0.03}$	$\mathbf{0.85 \pm 0.02}$	–	$\mathbf{35 \pm 2}\%$	–	$2.13\times$ batch

make conformal wrapping a practical companion to deterministic allocation scores in non-stationary deployment.

I Detailed Ablation Results

In the matched EM experiments, entropy is stronger than the alternative internal summaries tested. On EM, allocation entropy achieves $\rho_s = 0.67 \pm 0.03$, AUROC 0.82 ± 0.02, and Sel.@80% $31 \pm 2\%$. Alternative statistics are weaker: $\mathrm{Var}(g)$ $(0.59/0.76/25\%)$, Gini $(0.61/0.78/27\%)$, $\max(g)$ $(0.53/0.72/21\%)$, and feature norm $\|v\|$ $(0.56/0.74/23\%)$. This indicates that uncertainty is carried by allocation ambiguity itself rather than by activation magnitude alone.

In the same control suite, excess entropy H_E is kept as an anti-confound diagnostic rather than a replacement main-text score: residualizing against input-only complexity reduces magnitude, which is expected if part of the raw signal tracks geometric difficulty, but the remaining allocation-specific component stays useful as a diagnostic check against the "hard inputs only" interpretation. At similar cost, allocation entropy is also stronger than the efficient low-cost ensemble variants in the matched EM protocol. Snapshot [35] (3), Multi-Head (3), and Batch Ensemble [36] reach $\rho_s = 0.59/0.61/0.58$ with AUROC $0.77/0.78/0.76$ at $1.5\times$/$1.2\times$/$1.3\times$ cost, while H_L keeps $\rho_s = 0.67$ and AUROC 0.82 at $1.13\times$ batched cost. Representation dependence is explicit: the same WNO-style local rule does not preserve the multi-scale frequency structure needed for a fair Fourier uncertainty benchmark, so the adapter target is the frequency-band construction in Table 8 rather than raw Fourier coefficients. The released code implements that adapter path, but this paper does not present a matched FNO uncertainty benchmark. The same caution applies to skip-connected backbones: a naive U-WNO+gating port collapses to test relative L^2 22.43 versus 0.535 for vanilla U-WNO in the recorded control run, so broader architecture transfer requires adapter-specific design rather than copying the local wavelet gate unchanged. Across Haar, db2, db4, and Coiflet-1, ranking quality remains in the $\rho_s = 0.67$–0.69 range.

J Cross-Domain Diagnostics and Scaling Notes

On PDEBench, transfer quality is task-dependent. Darcy exhibits the strongest ranking signal; Navier-Stokes is harder due to mixed local/global dynamics. Table 11 reports the compact summary; the full tuned-UQ baseline comparison is reported separately in the supporting follow-up suite.

The stress matrix (Table 12) shows prediction robustness of wavelet models across shift families. It does not by itself validate allocation entropy; ranking, shifted AUROC, and conformal coverage require matched stress splits where those diagnostics are available.

Table 11: PDEBench transfer summary. "Pred. error" is relative L^2; FNO error is the available prediction baseline.

Task	Pred. err.↓	FNO err.↓	ρ_s ↑	AUROC↑
Time-dependent and cross-PDE stress tasks				
Darcy flow	$\mathbf{0.0298 \pm 0.0012}$	$\mathbf{0.0391}$	$\mathbf{0.70 \pm 0.03}$	$\mathbf{0.84 \pm 0.02}$
Diff.-reaction	0.0382 ± 0.0015	0.0467	0.65 ± 0.04	0.80 ± 0.03
Comp. NS 2D	0.0421 ± 0.0018	0.0538	0.63 ± 0.04	0.79 ± 0.03

Table 12: Prediction-only cross-domain stress matrix from the existing stress report (multi-seed mean±95% CI, relative L^2 error in %). This table supports representation robustness but is not used as evidence for uncertainty calibration or OOD ranking.

Shift setup	MWNO-AUD	WNO	FNO
In-domain (mid→mid)	$\mathbf{3.08 \pm 0.19}$	3.12 ± 0.21	5.23 ± 0.34
Frequency OOD (mid→high)	$\mathbf{4.51 \pm 0.37}$	5.08 ± 0.42	8.92 ± 0.71
Geometry OOD (simple→mixed)	$\mathbf{3.79 \pm 0.28}$	4.41 ± 0.36	7.28 ± 0.59
Material OOD (mid→high)	$\mathbf{4.18 \pm 0.34}$	4.87 ± 0.41	8.14 ± 0.68
Combined stress (3-way)	$\mathbf{6.42 \pm 0.58}$	7.83 ± 0.71	13.51 ± 1.12

J.1 Resolution Scaling and Extreme Shift Commentary

At 128×128, the experiments report improved prediction error (relative L^2: 0.5412 ± 0.0021 vs 0.5893 ± 0.0019 at 64×64) while preserving low overhead ($1.10\times$). UQ quality drops mildly (ρ_s: 0.67→0.64, AUROC: 0.82→0.80), showing graceful scaling within this setting.

For combined frequency+geometry+material shift, MWNO-AUD degrades less than FNO (error multiplier $1.55\times$ vs $2.23\times$). This is consistent with locality: wavelet coefficients can re-concentrate near newly shifted interfaces, while global Fourier representations require broader reallocation. The existing stress report supports MWNO-AUD prediction robustness under this split, while UQ usefulness under the same stress test is treated as a separate uncertainty diagnostic.

J.2 PDEBench Transfer and Deployment Regimes

Across PDEBench tasks, allocation entropy remains useful but task-dependent (Darcy/diffusion-reaction/NS: $\rho_s = 0.70/0.65/0.63$; AUROC $0.84/0.80/0.79$). Table 8 documents adapter coverage in the code, but architecture-transfer evidence remains uneven across families. Under combined shift, MWNO-AUD degrades more gracefully than WNO/FNO in prediction error; allocation-entropy usefulness for staged escalation is evaluated on the matching shifted splits when uncertainty labels are available. We summarize deployment regimes as mild (H_L only), moderate (top 10–20% escalation), and severe (hybrid with small ensembles).

Closed-loop deployment is outside the main benchmark: allocation-priority active learning and drift monitoring are treated as deployment stress tests against random, ensemble-disagreement, and oracle acquisition under matched simulator budgets. The streaming-escalation setup, including static-threshold versus entropy-quantile comparison and budget sensitivity, is specified in Appendix L.

K Spatial Mechanism Visual Diagnostics

Figures 3 and 5 illustrate the spatial diagnostics used to inspect the mechanism. The relevant quantitative check is whether error hotspots and high-allocation-entropy regions overlap around material interfaces and diffraction boundaries. The mechanism suite also includes active-set flips, residual-overlap diagnostics, excess-entropy controls, random/frozen gate controls, capacity sweeps, and training trajectories. These controls are the empirical bridge for Proposition 3: they test whether entropy rises with active-set instability and whether the resulting residual remains concentrated in the same localized regions where the wavelet basis is expected to be sparse and physically meaningful. In this submission, we use Figures 3 and 5 together with those controls as supporting diagnostics rather than as standalone quantitative proof of localization.

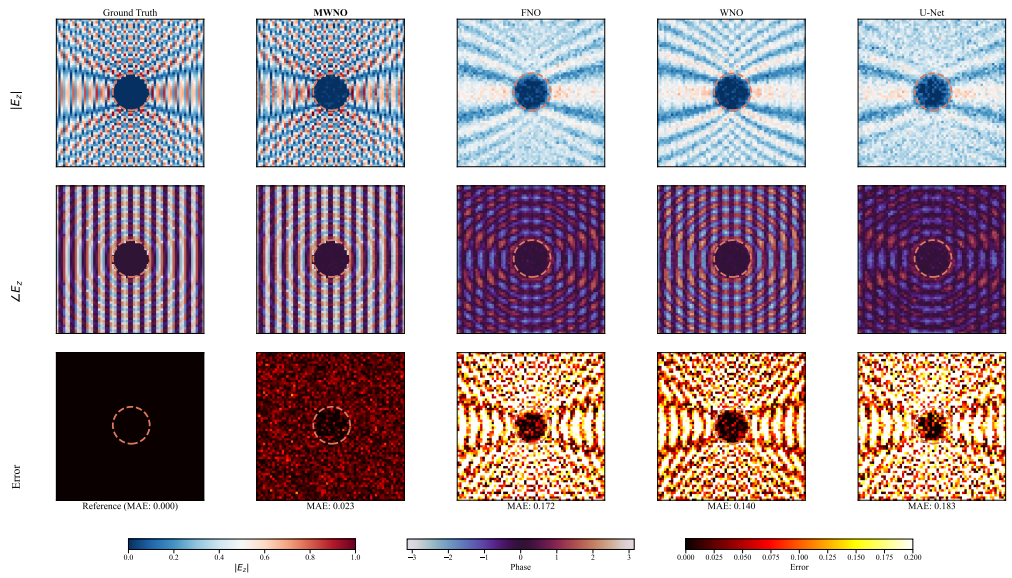


Figure 5: Prediction and error maps on a representative EM sample. Columns compare ground truth/reference fields with MWNO-AUD, FNO, WNO, and U-Net outputs; rows show field magnitude, phase, and pointwise error. Dashed circles mark the scatterer boundary, and the error row uses a perceptually uniform color scale for grayscale and colorblind readability.

L Streaming Escalation Case Study

The deployment motivation for allocation entropy is streaming triage: use the single-pass score to decide which samples require expensive fallback evaluations, ensemble escalation, or calibration refresh. Closed-loop deployment remains outside the main benchmark, so this appendix describes the stress protocol rather than core reported numbers. The streaming study is informative for budget sensitivity and policy wiring, but remains supporting deployment evidence rather than the final closed-loop claim. The case study compares entropy-priority acquisition, random acquisition, ensemble-disagreement acquisition, and oracle acquisition under the same simulator budget. It reports drift-detection delay, cumulative risk under static versus quantile thresholds, and sensitivity to the escalation fraction q within that stress-test setting. The purpose of this case study is to document how the score can be wired into staged deployment, not to elevate closed-loop control to the paper's main benchmark.

M Broader Impact and Responsible Use

The intended positive impact of this work is narrower and more practical than a general-purpose uncertainty framework. Scientific-surrogate deployments often waste compute by treating every query identically even when only a small subset requires expensive fallback evaluation. A useful single-pass triage score can reduce unnecessary solver or ensemble calls, improve screening efficiency in simulation-heavy workflows, and make uncertainty-aware deployment more accessible in resource-constrained scientific settings.

The main misuse risk is over-reading the score as a calibrated posterior or a stand-alone safety certificate. AUD is not designed for that role, and the paper explicitly shows regimes in which the local wavelet score weakens. For transport-heavy, long-horizon, volumetric, or otherwise mismatched regimes, the responsible use pattern is held-out recalibration plus escalation to stronger global models, ensembles, or solvers rather than relying on raw entropy alone. The anonymized supplementary code package therefore documents the audited reproduction path, smoke validation workflow, and asset notes together with the code and experiment configurations used to support the main deployment claim.

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